

Text to Image Generation

```
! pip install -q diffusers transformers accelerate
```

```
from diffusers import StableDiffusionPipeline
```

```
pipe = StableDiffusionPipeline.from_pretrained(  
    "runwayml/stable-diffusion-v1-5"  
)
```

```
prompt = input("Enter prompt: ")
```

```
image = pipe(prompt).images[0]
```

```
image.save("output.png")  
image.show()
```

Text to Text Language Translation

! pip install translate

```
from translate import Translator
```

```
text = input("Enter text: ")
```

```
lang = input("Enter language code: ")
```

```
translator = Translator(to_lang=lang)
```

```
print("Translation:", translator.translate(text))
```

Text to Music Generation

```
! pip install -q transformers scipy
```

```
from transformers import AutoProcessor,  
MusicgenForConditionalGeneration  
from IPython.display import Audio
```

```
p = AutoProcessor.from_pretrained("facebook/musicgen-small")  
m =  
MusicgenForConditionalGeneration.from_pretrained("facebook/musicgen-s  
mall")
```

```
x = p(text="happy piano music", return_tensors="pt")  
y = m.generate(**x, max_new_tokens=256)
```

```
Audio(y[0].numpy(), rate=32000)
```

Image to Text Generation

```
!pip install -q transformers pillow
```

```
from transformers import pipeline  
from PIL import Image
```

```
captioner = pipeline(  
    "image-text-to-text",  
    model="Salesforce/blip-image-captioning-base"  
)
```

```
image = Image.open("/content/mountain.jpg")
```

```
result = captioner(  
    images=image,  
    prompt="Describe this image."  
)
```

```
print(result)
```

Text to Speech Generation

```
!pip install -q gtts
```

```
from gtts import gTTS  
from IPython.display import Audio
```

```
text = "Hello, this is my generated music"  
tts = gTTS(text)  
tts.save("music.mp3")
```

```
Audio("music.mp3")
```

Text to Story Generation

```
!pip install -q transformers
```

```
from transformers import pipeline
```

```
story = pipeline("text-generation", model="gpt2")
```

```
prompt = input("Enter story starting line: ")
```

```
result = story(  
    prompt,  
    max_new_tokens=100  
)
```

```
print(result[0]["generated_text"])
```

AI Chat Bot

```
!pip install -q transformers accelerate
```

```
from transformers import pipeline
```

```
bot = pipeline(  
    "text-generation",  
    model="Qwen/Qwen2.5-0.5B-Instruct"  
)
```

```
while True:
```

```
    q = input("You: ")
```

```
    if q.lower() in ["exit", "quit", "bye"]:  
        break
```

```
    messages = [{"role": "user", "content": q}]
```

```
    result = bot(messages, max_new_tokens=100)
```

```
    print("Bot:", result[0]["generated_text"][-1]["content"])
```

Image to Sketch Generation

! pip install opencv-python

```
import cv2
from google.colab.patches import cv2_imshow

img = cv2.imread("/content/image.jpg")

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

blur = cv2.GaussianBlur(255 - gray, (21, 21), 0)

sketch = cv2.divide(gray, 255 - blur, scale=256)

cv2_imshow(sketch)
```


Music to Text Generation

```
!pip install -q openai-whisper
```

```
import whisper
```

```
model = whisper.load_model("base")
```

```
audio = "/content/music.mp3"
```

```
result = model.transcribe(audio)
```

```
print(result["text"])
```

```
open("music_text.txt", "w").write(result["text"])
```